Attachment Style Prediction Using Machine Learning

Combining structured psychological data with AI-generated relationship reflections

Executive Summary

- A machine learning model was developed to predict four attachment styles using structured survey data and AI-generated text reflections
- The final model combined psychological survey features with text embeddings generated from synthetic first-person reflections
- A tuned XGBoost model delivered the strongest overall performance with approximately 58% accuracy and 57% macro F1 score
- The Fearful Avoidant attachment style was identified most accurately by the model
- While further validation using real-world user text is recommended, the results demonstrate encouraging potential for applying AI-driven behavioural prediction within commercial relationship platforms

Why Attachment Style Prediction Matters

- Attachment styles influence communication patterns, emotional regulation and relationship behaviour
- Dating and relationship platforms could use predictive models to personalise experiences and improve compatibility recommendations
- Early identification of attachment tendencies may support safer and healthier user interactions
- The project demonstrates a potential bridge between behavioural psychology and applied AI systems

Potential Applications

- **Dating app recommendations**
- **Relationship coaching tools**
- **Personalised onboarding journeys**
- **Behaviour segmentation**
- **Mental wellbeing support**

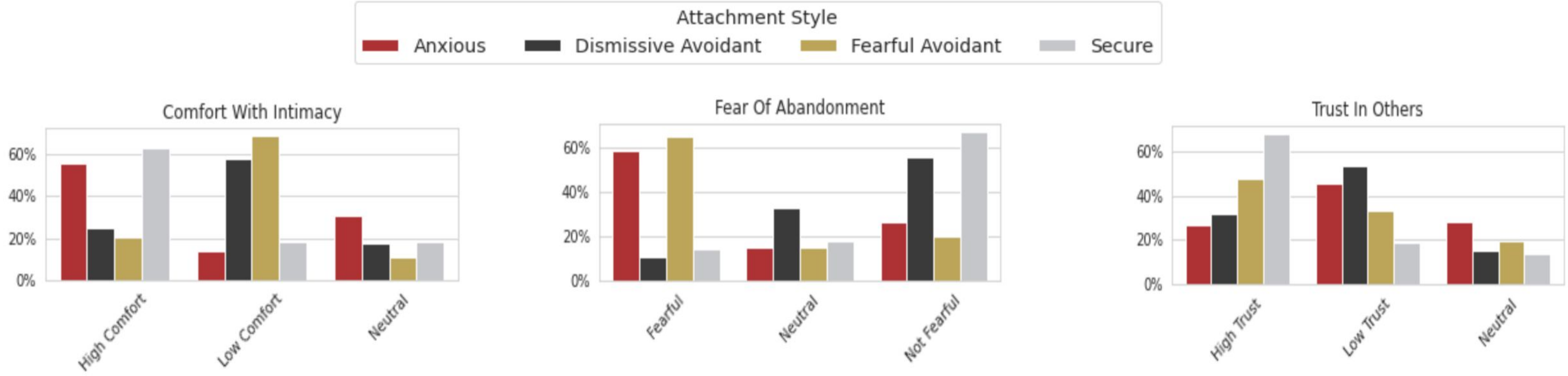
Generating Synthetic Relationship Reflections

- The original dataset contained structured survey responses but no free text data
- OpenAI was used to generate short first-person relationship reflections for each user
- Prompts were constructed only from structured input features and intentionally excluded the attachment style label to avoid leakage
- The generated text aimed to simulate how users may naturally describe relationship experiences and emotions
- These reflections were converted into embeddings using the sentence-transformers/all-MiniLM-L6-v2 model

Example generated reflection:
‘I sometimes worry that people will leave me and I often need reassurance in relationships. I value closeness but can become anxious when communication changes suddenly’

Exploratory Analysis: Key Findings

- The dataset was skewed towards avoidant attachment styles which may introduce modelling bias
- Core psychological variables relating to trust, abandonment fears and emotional intimacy showed the strongest relationship with attachment style
- Psychological and behavioural indicators were substantially more correlated with attachment style than demographic-style variables
- The analysis suggested that attachment styles can be differentiated using a combination of emotional and behavioural traits



Modelling Approach

- Three modelling approaches were evaluated:
  1) Structured survey data only
  2) Text embeddings only
  3) Combined structured + text embeddings

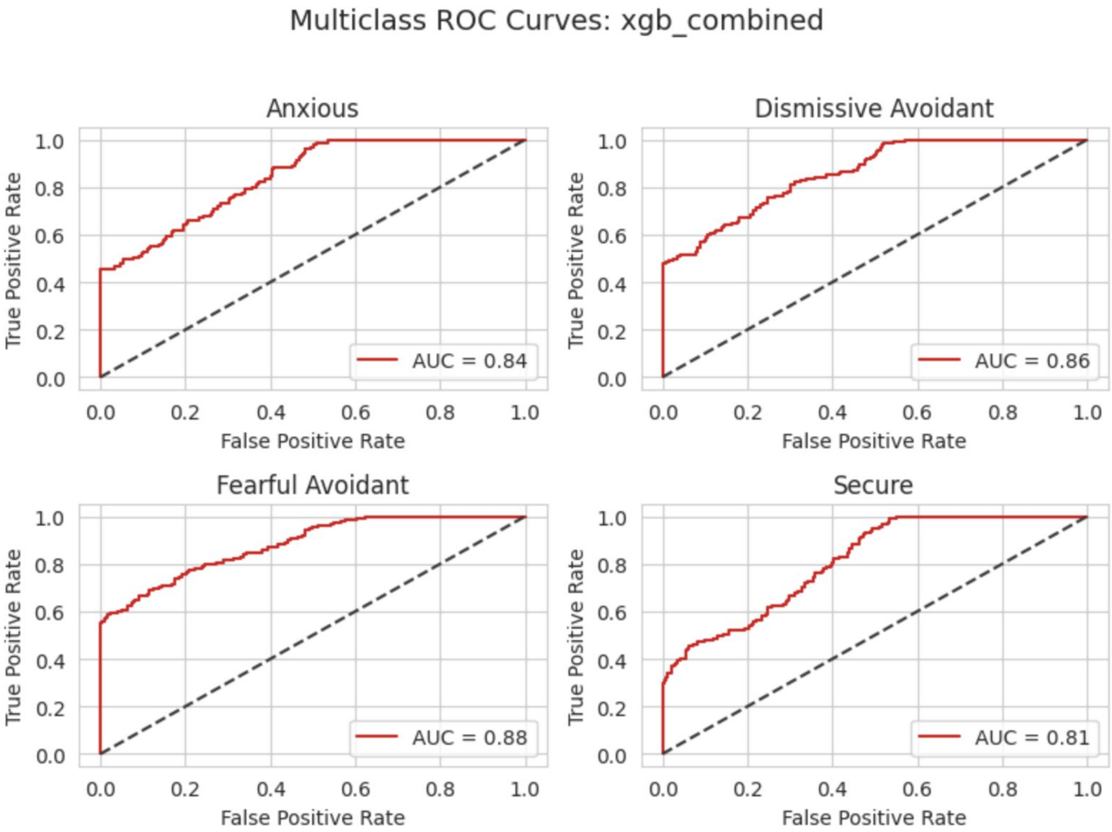
- Models tested included Logistic Regression, Random Forest and XGBoost

- Train/test split was performed before embedding generation to avoid leakage

- Hyperparameter tuning was carried out using RandomizedSearchCV

Approach	Strength	Weakness	Outcome
Structured	Strong baseline	Limited nuance	Performed well
Embeddings	Captured language	Weak alone	Poor overall
Combined	Best balance	More complex	Best final model

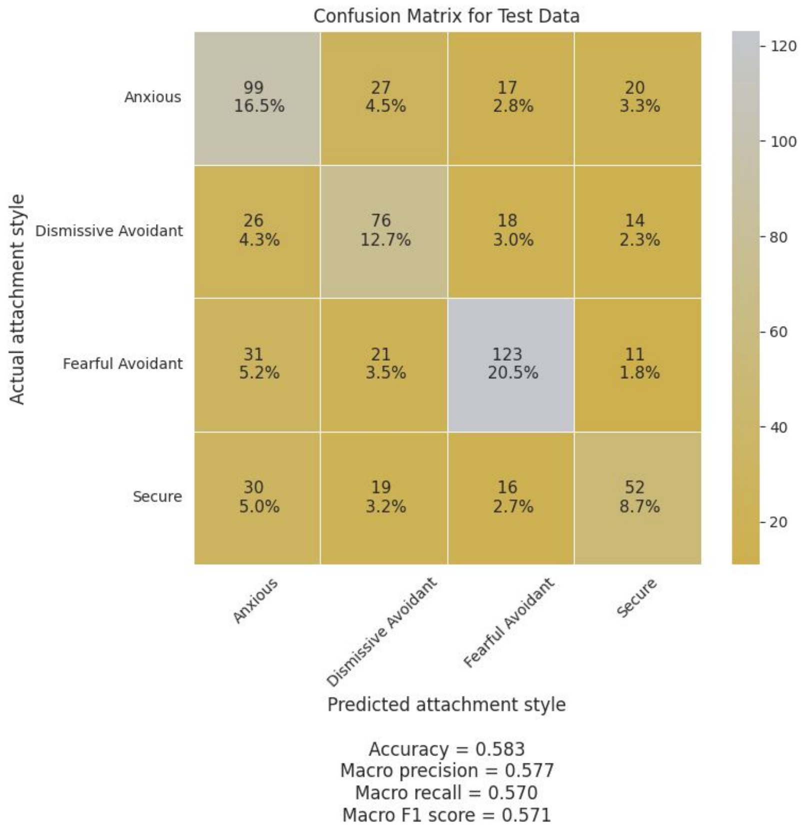
Model Performance: ROC Curves



Key observations

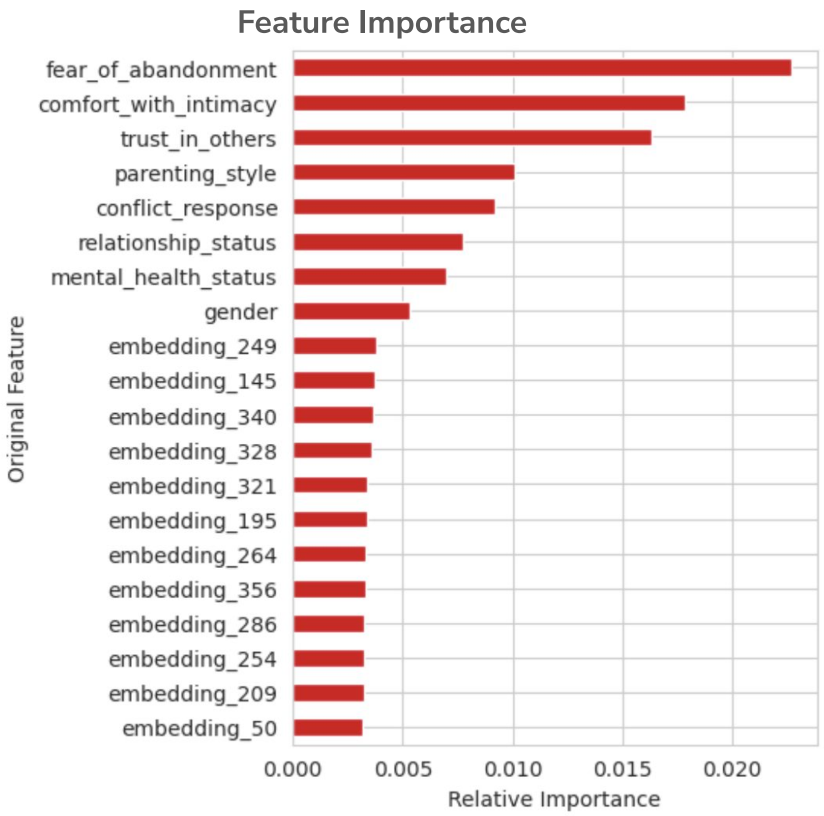
- Combined model outperformed embeddings-only models
- Fearful Avoidant class achieved the strongest discrimination
- Secure attachment style remained hardest to identify consistently
- ROC curves demonstrate stable multiclass separation across classes

Final Model Evaluation: Confusion Matrix



- Diagonal values represent correct predictions
- Fearful Avoidant users were predicted most accurately
- Secure users were the most difficult for the model to predict
- Overall performance remained balanced across all four classes

What Drove Predictions?



Key drivers

- Fear of abandonment
- Comfort with intimacy
- Trust in others
- Parenting style
- Conflict response
- The structured features were more predictive than the text embeddings

Final Results and Interpretation

Final XGBoost Model

Accuracy: ~58%
Macro F1*: ~57%
Multiclass AUC: ~0.84

Interpretation

- Text embeddings provided incremental value when combined with structured features
- Random Forest performance degraded slightly after adding embeddings
- XGBoost handled the combined feature space more effectively
- Results are encouraging given the complexity and subjectivity of psychological traits

* Macro F1 was used because it evaluates how consistently the model performs across all attachment styles rather than rewarding strong performance on only the most common groups

Recommendations and Next Steps

- Operationalise the tuned XGBoost model through an Airflow pipeline, enabling scheduled attachment-style predictions to be written to SQL and consumed by analytics, dashboards or product systems
- Collect real-world free text data rather than relying on synthetic reflections
- Expand the dataset to improve representation of underrepresented attachment styles
- Test transformer-based architectures for improved text understanding
- Explore applications in dating apps, relationship coaching and behavioural personalisation systems
- Introduce explainability tooling to help interpret model decisions in production

Key Takeaway

Combining structured psychology data with language-based AI signals can create richer behavioural prediction systems with real-world commercial applications